

Tic Tac Toe Game (AI)

Aim

To implement the Tic Tac Toe game and simulate a sequence of moves between two players, checking for a winning condition after every move.

Objectives

- To represent the 3x3 game board using a data structure.
- To implement turn-based play between two players (X and O).
- To check rows, columns and diagonals for a winning condition.
- To correctly declare a winner or a draw.

Algorithm

- 1 Start.
- 2 Initialize an empty 3x3 board.
- 3 Player X and Player O take turns marking empty cells.
- 4 After each move, update the board and display it.
- 5 Check all rows, columns, and diagonals for three identical marks.
- 6 If a winning combination is found, declare that player the winner and stop.
- 7 If the board is full with no winner, declare the game a draw.
- 8 Repeat until a winner is found or the board is full.
- 9 Stop.

Source Code (Python)

```
board = [' '] * 9
def print_board():
    for i in range(0, 9, 3):
        print(f" {board[i]} | {board[i+1]} | {board[i+2]} ")
        if i < 6:
            print("---+---+---")
def check_winner():
    wins = [(0,1,2),(3,4,5),(6,7,8),(0,3,6),(1,4,7),(2,5,8),(0,4,8),(2,4,6)]
    for a, b, c in wins:
        if board[a] != ' ' and board[a] == board[b] == board[c]:
            return board[a]
    if ' ' not in board:
        return 'Draw'
    return None
# Simulated sequence of moves alternating between Player X and Player O
moves = [(0, 'X'), (4, 'O'), (1, 'X'), (5, 'O'), (2, 'X')]
print("Tic Tac Toe - Game Simulation\n")
for pos, player in moves:
    board[pos] = player
    print(f"Player {player} plays position {pos}:")
    print_board()
    result = check_winner()
    if result:
```

```
        print(f"\nResult: {'Draw' if result == 'Draw' else f'Player {result} wins!'}")
        break
    print()
```

Output

```
Tic Tac Toe - Game Simulation
Player X plays position 0:
X |  | 
---+---+---
  |  | 
---+---+---
  |  | 
Player O plays position 4:
X |  | 
---+---+---
  | O | 
---+---+---
  |  | 
Player X plays position 1:
X | X | 
---+---+---
  | O | 
---+---+---
  |  | 
Player O plays position 5:
X | X | 
---+---+---
  | O | O
---+---+---
  |  | 
Player X plays position 2:
X | X | X
---+---+---
  | O | O
---+---+---
  |  | 
Result: Player X wins!
```

Result

The Tic Tac Toe game was successfully simulated. Player X completed the top row (positions 0, 1, 2) and was declared the winner.